## Exercises

**Exercise 11.1 · From the graph to the matrix.** A probabilistic graph has three vertices 1, 2, 3 and the following arcs: $1 \to 1$ ($\frac{1}{2}$), $1 \to 2$ ($\frac{1}{4}$), $1 \to 3$ ($\frac{1}{4}$), $2 \to 1$ ($\frac{1}{2}$), $2 \to 3$ ($\frac{1}{2}$), $3 \to 1$ ($\frac{1}{4}$), $3 \to 2$ ($\frac{1}{4}$), $3 \to 3$ ($\frac{1}{2}$).

1. Write the transition matrix $T$ and check the row sums.
2. Starting from state 1, compute the distribution after two transitions.

**Exercise 11.2 · Two network operators.** Each month, 10% of the customers of operator $A$ move to $B$ and 20% of the customers of $B$ move to $A$; the others stay. At the start, each operator has half of the customers. Let $x_n$ be the share of $A$ after $n$ months.

1. Write the transition matrix and find the invariant distribution.
2. Prove that $x_{n+1} = 0.7x_n + 0.2$, then that $x_n = \frac{2}{3} - \frac{1}{6} \times 0.7^n$.
3. Find the smallest $n$ such that $x_n > 0.66$.

**Exercise 11.3 · A three-state walk.** A particle moves on three squares in a row, 1, 2, 3. From square 2, it goes to 1 or to 3 with probability $\frac{1}{4}$ each and stays on 2 with probability $\frac{1}{2}$. From an end square, it stays where it is or goes to 2, with probability $\frac{1}{2}$ each.

1. Write the transition matrix $T$.
2. Check that $\begin{pmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{pmatrix}$ is an invariant distribution.
3. Prove that it is the only one.

**Exercise 11.4 · Two tanks.** The amount of water (in thousands of litres) in two connected tanks is modelled by $U_n = \begin{pmatrix} x_n \\ y_n \end{pmatrix}$ with

$$U_{n+1} = AU_n + B, \qquad A = \begin{pmatrix} 0.6 & 0.2 \\ 0.1 & 0.7 \end{pmatrix}, \qquad B = \begin{pmatrix} 10 \\ 5 \end{pmatrix}.$$

1. Find the column $C$ such that $C = AC + B$.
2. Express $U_n$ in terms of $A^n$, $U_0$ and $C$.
3. We take for granted that the entries of $A^n$ tend to 0. Find the limit of $U_n$. Does it depend on $U_0$?

**Exercise 11.5 · Transitions in three steps.** Each day, a user of an app is active (1) or inactive (2), with transition matrix $T = \begin{pmatrix} 0.7 & 0.3 \\ 0.5 & 0.5 \end{pmatrix}$.

1. Compute $T^2$ and $T^3$.
2. A user is active on day 0. What is the probability that they are inactive on day 3?
3. Find the invariant distribution.

**Exercise 11.6 · A no-claims bonus system.** Each year an insurer classes its customers in three levels: bonus (1), standard (2), penalty (3). The annual transition matrix is

$$T = \begin{pmatrix} 0.8 & 0.2 & 0 \\ 0.6 & 0.2 & 0.2 \\ 0 & 0.6 & 0.4 \end{pmatrix}.$$

1. Draw the probabilistic graph.
2. Find the invariant distribution.
3. A customer is at the standard level. What is the probability of being at the bonus level two years later?

**Exercise 11.7 · Powers of the two-state transition matrix.** Let $T = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}$ with $a+b \neq 0$. Let $\lambda = 1-a-b$, $Q = \frac{1}{a+b}\begin{pmatrix} b & a \\ b & a \end{pmatrix}$ and $R = \frac{1}{a+b}\begin{pmatrix} a & -a \\ -b & b \end{pmatrix}$.

1. Check that $Q + \lambda R = T$, $Q^2 = Q$, $R^2 = R$ and $QR = RQ = 0$.
2. Deduce by induction that $T^n = Q + \lambda^n R$ for every $n \geq 1$.
3. Assume $0 < a + b < 2$. Find the limit of $T^n$ and interpret its rows.